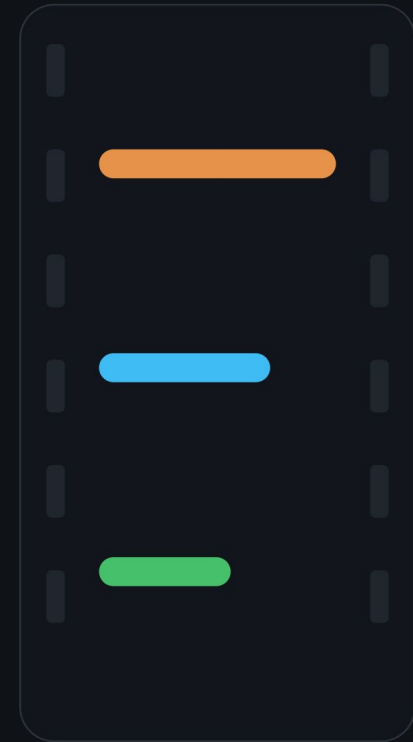


AI CONSULTING CAPSTONE · ROUND 2

Taste Agent for Letterboxd

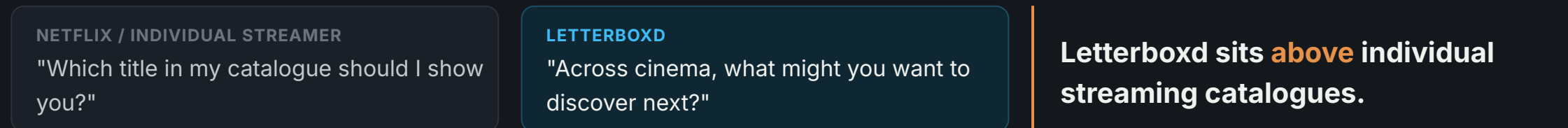
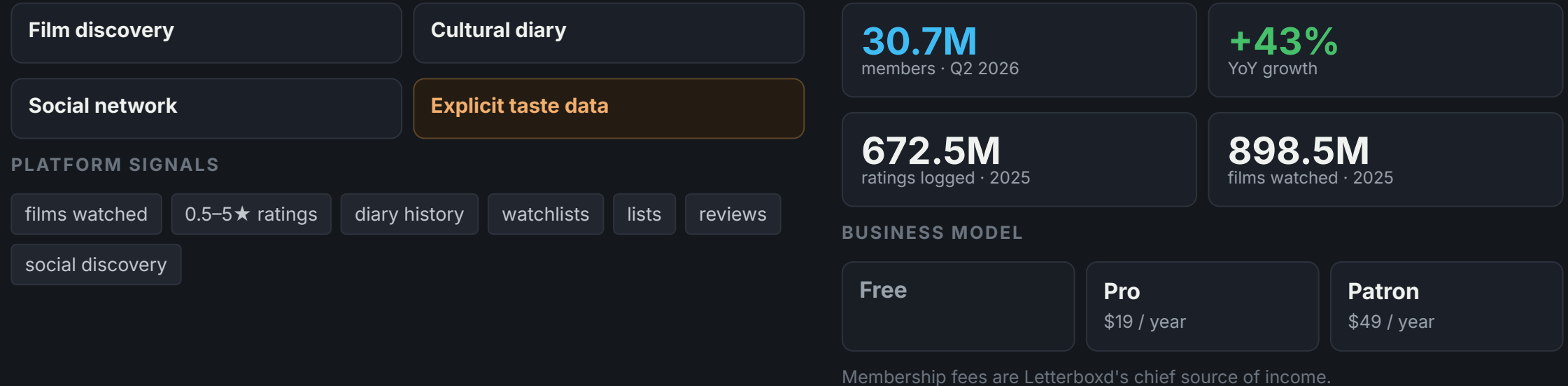
From film history to contextual cultural discovery

— Marc Tanguy — AI Consulting Capstone



Letterboxd is not a streaming service.

That is exactly why this opportunity exists.



Sources: Tiny Q2 2026 results · Letterboxd 2025 Year in Review · Letterboxd Purpose / Pro pages. Different product problem from Netflix — not a weaker one.

The opportunity is bigger than "another recommender".

CULTURAL TASTE INTELLIGENCE



WHY THIS MATTERS

Not "AI recommends movies."

Letterboxd can turn a user's cultural history into an **interpretable taste layer** that supports multiple discovery experiences.

- My Taste
- Taste Evolution
- Semantic Search
- Taste Agent
- Personalized Discovery

A differentiated product proposition.

✓ **Cross-media aggregation** — demonstrated capability

● **Cross-media predictive lift** — still unproven

We proved film and book histories share one semantic architecture — not that book data improves film-ranking accuracy.

I did not protect the original hypothesis. I tested it.

ROUND 1 POC

107

works · 96 films + 11 books

62

semantic dimensions

n8n → enrichment → semantic classification → Taste Model

- ✓ semantic representation feasible
- ✓ films + books, same architecture
- ✓ structured, explainable profile

PEER CHALLENGE

"Does this actually produce better recommendations?"

"Why is this different from Netflix?"

"Could the profile become inaccurate or overly personal?"

"Focus the MVP on movies first."

ROUND 2 FINDINGS

- 1 · The 62D semantic model did not beat the simpler metadata baseline as a film-preference predictor.
- 2 · Goodreads showed no detectable incremental predictive value for film ranking.
- 3 · History alone introduced selection bias — it fits films already chosen, not current unseen-film intent.

HOW THE MVP RESPONDED

SEMANTIC EMBEDDING LAYER

✓ reassigned

~1K film embeddings

Dense representations for retrieval & candidate qualification

OpenAI embeddings · 1536D



HISTORICAL PREFERENCE MODEL

✓ reassigned

ML compatibility model

Learns compatibility from rated history — applied after relevance

PCA + RidgeCV · frozen history



INTERPRETABLE TASTE MODEL

✓ reassigned

62D statistical Taste Profile

Pearson associations across theme, tone, pacing, style, narrative

62 dimensions · explanation

Components of the evolved MVP — not one sequential score



So I changed the architecture.

KEEP

Taste Profile for interpretation, aggregation, explanation

DON'T

Use it as the sole recommendation engine

BECOMES

Contextual film discovery = request relevance + watchlist intent + historical compatibility

One profile. A different engine for the decision.

LAYER 1 · CULTURAL TASTE PROFILE

Understand, make visible, explain

Letterboxd history

Goodreads history

→

shared semantic representation

→

interpretable Taste Profile

Does not independently determine the recommendation universe.

LAYER 2 · CONTEXTUAL FILM DISCOVERY

Decide the recommendation

current request

watchlist

history compatibility

semantic evidence

User request

→

understand intent

→

retrieve films

→

hard constraints

→

qualify candidates

→

personalize (history)

→

watchlist intent

→

select

→

explain from evidence

→

validate / abstain

| Current intent establishes relevance. Historical taste personalizes second.

MVP DEMO

From "my taste" to "what should I watch tonight?"

1 · MY TASTE

Historical cultural behaviour → interpretable taste signals



2 · ASK TASTE AGENT

"I want something English from the 80s."



3 · CONTEXTUAL DISCOVERY

Respects the request, excludes watched, flags watchlist films

Ask Taste Agent →

Maurice
1987

The Woman in Black
1989

The Hunger
1983

Already on your radar

Miss Marple
1989

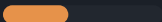
The Shining
1980

| This is what the POC could not prove: the semantic layer is now connected to a real user decision.

The MVP works. The evidence tells us where not to overclaim.

MODEL LEARNING

Metadata  0.319

62D  0.130

Spearman · frozen holdout

Semantic profiling ≠ better ranking by itself.

HUMAN REVIEW

84

unique prompt–film ratings · 4
conditions · 48 rows

One evaluator / limited sample.

Useful product evidence, not final
market validation.

PERFORMANCE

160.72s → 48.35s

–69.9% latency · ~3.3× faster

median runtime

Big engineering gain — still not
consumer–production latency.

QUALITY / SAFETY

- ✓ hard constraints
- ✓ watched–film exclusion
- ✓ grounded explanations
- ✓ valid abstention / partial slate

| The remaining question is **user value at Pilot scale** — not technical feasibility.

The investment is measurable. The return still needs Pilot validation.

COST — MARKET-ANCHORED (MALT)

€102K
one-time build

€34K
annual operation

€136K
Year 1

€204K
36 months

AI / ML eng

Full-stack

MLOps

UX

QA

Security

PM

privacy / legal

VALUE MODEL — BASE CASE

≈ 43%
12-month ROI

≈ 185%
36-month ROI

Value drivers:

Free → paid conversion

paid retention

Pro → Patron upgrades

discovery engagement

Conservative

Year-1 ROI can be < 0%

Base

≈ +43% at 12 months

Upside

≈ +136% at 12 months

A conservative case that can go negative is **why we recommend a Pilot** — not full funding.

Scenario model — not a Letterboxd forecast. Cost anchored to public Malt category day-rate benchmarks; no individual freelancers named.

Personalization only works if users can trust it.

1 · PRODUCT RISK

Recommendation quality

An interesting profile does not necessarily mean useful recommendations.

Mitigation · baseline comparison + real-user Pilot

2 · TECHNICAL RISK

Latency & external AI/API dependencies

Current: **48.35s** median MVP runtime.

Mitigation · precomputation, caching, production retrieval, SLO during Pilot

3 · PRIVACY / AI GOVERNANCE

EU AI Act · entertainment recommendation is not high-risk under the intended purpose; user-facing AI transparency still applies.

GDPR · ratings, history, watchlists and inferred Taste Profiles are personal data.

lawful basis

purpose limitation

minimisation

correction / deletion

retention policy

processor / transfer review

DPIA before production

**Model cultural preference
not personal identity**

DO NOT INFER

politics

religion

health

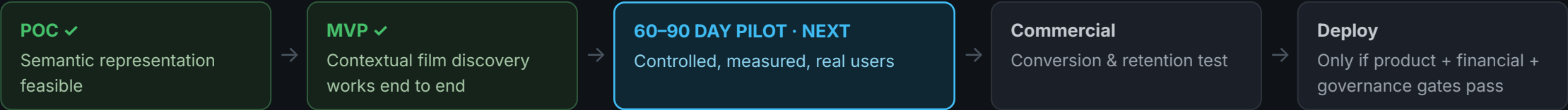
sexual orientation

ethnicity

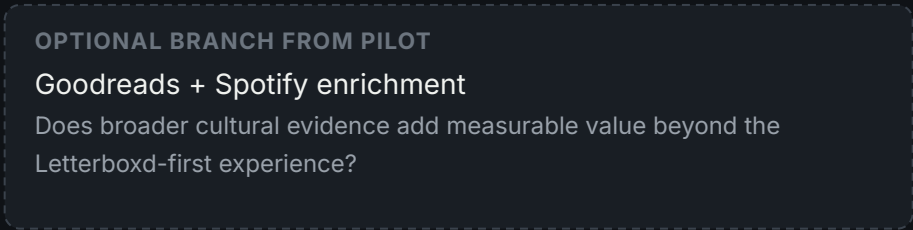
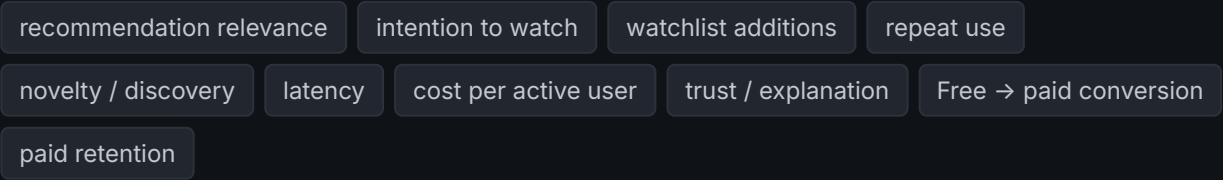
personality

RECOMMENDATION

The next decision is not "deploy or stop". It is "does this earn a Pilot?"



THE PILOT MEASURES



Approve a 60–90 day controlled Pilot.

"Letterboxd already knows what millions of people watch. Taste Agent asks whether that history can become a product that understands what they are looking for next."